

CHAPTER 2: MECHANICAL WAVES

At the end of this chapter, you'll be able to:

SUBTOPIC	LEARNING OUTCOMES
2.1	Mechanical waves a) Explain the mechanical waves and its formation. b) State the properties of transverse wave and longitudinal wave. c) Define amplitude, A, frequency, f, period T, wavelength, λ and wave number, k. d) Interpret and use equation for progressive wave, $y(x, t) = A \sin(\omega t \pm kx)$ e) State and use particle vibrational velocity as $v = \frac{dy}{dt}$ f) Use wave propagation velocity, $v = \lambda f$
2.2	Standing waves a) State the principle of superposition of waves for the constructive and destructive interferences. b) Use the standing wave equation, $y = 2A \cos kx \sin \omega t$ c) Compare between progressive waves and standing waves. d) Use wave speed in a stretched string, $v = \sqrt{\frac{T}{\mu}}$

2.1 Mechanical Waves

What is a mechanical wave?

Mechanical wave is defined as **disturbance that travels through particles of the medium to transfer the energy.**

Basically, it is any waves that move through any medium (solid, liquid or gas).

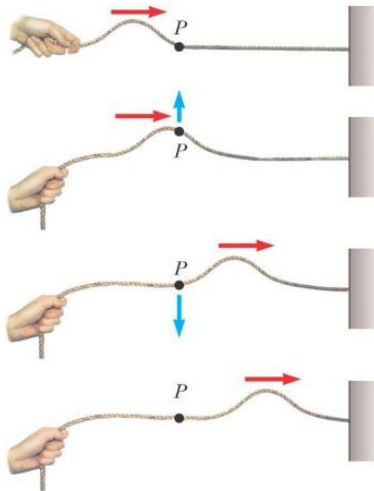
The **particles oscillate about their equilibrium position but do not travel.**

Examples: Water waves, sound waves, waves on a rope, waves in a spring.



Circular waves. Water waves move with velocity, but each particle (or molecule) of the water itself merely oscillate up and down about an equilibrium point.

Simulation:



Wave travelling on a rope. The wave travels to the right along the rope. Particles of the cord oscillate up and down.

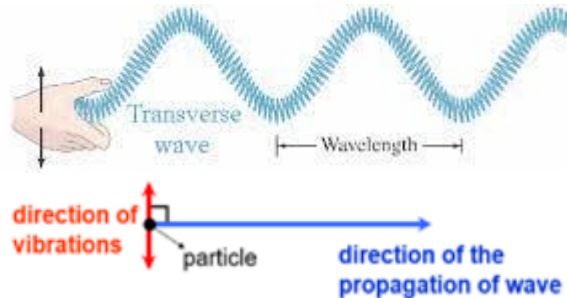
All mechanical waves require:

- Some **source of disturbance**
- A **medium** that can be disturbed
- A mechanism to transfer the disturbance from one point to the next point along the medium.

Two Types of Wave Motion

There are two types of wave motion:

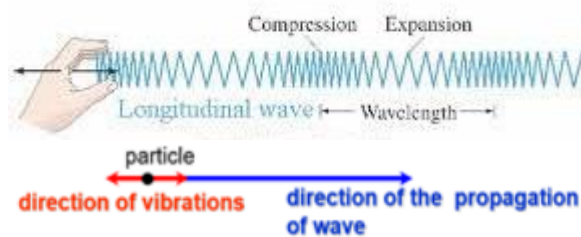
Transverse Wave



- A transverse wave is a wave in which the **particles** in the medium **vibrate perpendicular** to the direction which the wave propagates.
- So if the wave is travelling horizontally, the vibrations will be up and down.
- Example: Water waves, Waves on the rope/cord
- Simulation:



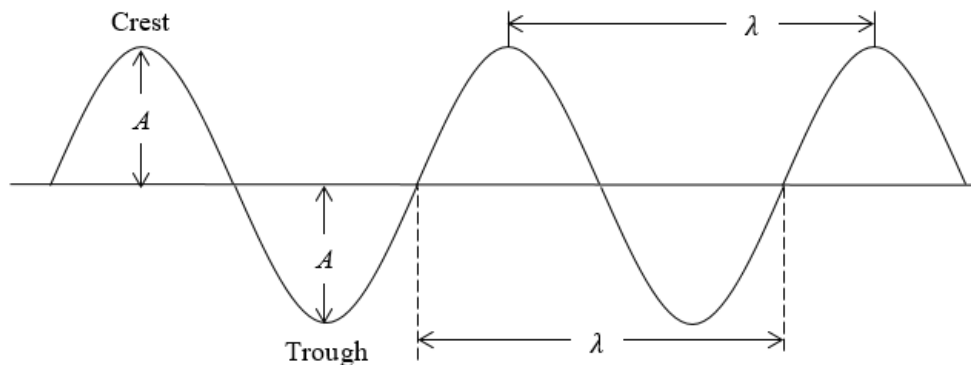
Longitudinal Wave



- A longitudinal wave is a wave in which the **particle** in the medium **vibrate parallel** to the direction which the wave propagates.
- So if the wave is travelling horizontally, the particles will be compressed closer together horizontally, or expanded horizontally as they go along.
- Example: Sound waves, waves on spring.
- Simulation:



Properties of Waves



Characteristic of a single-frequency continuous wave moving through space.

Quantities	Definition
Amplitude, A	Maximum displacement from the equilibrium position to the crest or trough of the wave motion.
Period, T	Time taken for the particle to complete one cycle of its oscillation. SI unit: s
Frequency, f	The number of complete cycles that pass a given point per second. $f = \frac{1}{T}$ SI unit: Hz
Wavelength, λ	The distance between two successive crests or trough of the wave. SI unit: m
Wave number, k	Number of waves or cycle per unit distance $k = \frac{2\pi}{\lambda}$ SI unit: m^{-1}

Equation for Progressive Wave (Travelling Wave)

Progressive wave is defined as wave that **propagated continuously outward** from a source of disturbance.

$$y(x, t) = A \sin(\omega t \pm kx)$$

y : distance move by a particle from its equilibrium position

x : distance of particle from the origin

A : amplitude of the wave

k : wave number, $k = \frac{2\pi}{\lambda}$

ω : angular frequency, $\omega = 2\pi f = \frac{2\pi}{T}$

+ or - : direction of wave propagation

The wave propagates to the left:

$$y(x, t) = A \sin(\omega t + kx)$$

The wave propagates to the right:

$$y(x, t) = A \sin(\omega t - kx)$$

Particle Vibrational Velocity and Wave Propagation Velocity

Particle Vibrational Velocity

Velocity of particle which **oscillate about equilibrium position** in simple harmonic motion that generate waves.

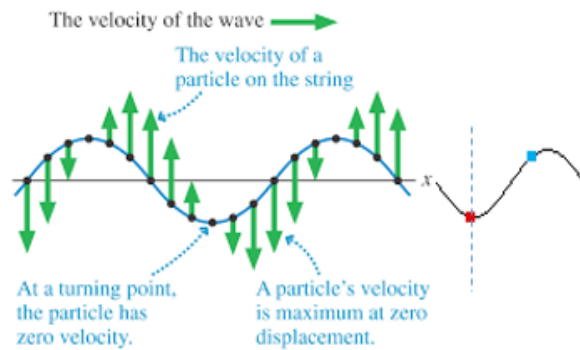
$$v_y = \frac{dy}{dt} = \frac{d}{dt}(A \sin(\omega t \pm kx))$$

$$v_y = A\omega \cos(\omega t \pm kx)$$

OR

$$v_y = \omega \sqrt{A^2 - y^2}$$

This velocity varies with time.



Wave Propagation Velocity

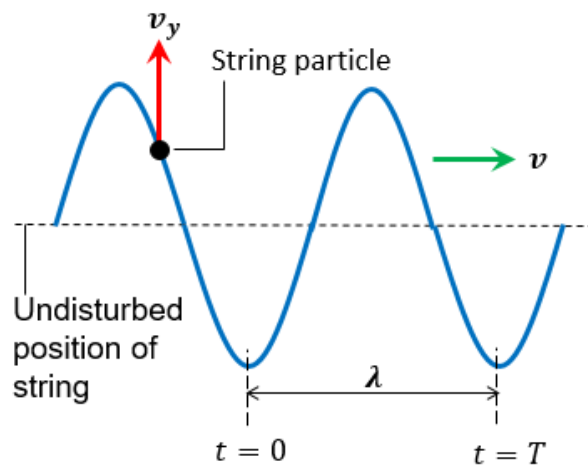
The wave velocity, v is the **velocity at which wave crests** (or other part of waveform) **move forward**.

Wave velocity is the **distance travel by wave profile per unit time**.

Figure shows the trough travels a distance of one wavelength, λ in a time equal to one period, T . Thus the wave velocity is:

$$v = \frac{\lambda}{T} \rightarrow v = f\lambda$$

This velocity does not varies with time (constant)



Example 1

A plane progressive wave is represented by the equation;

$$y = 20 \sin(2\pi t - 0.5\pi x)$$

Where x and y in meter and t is in second. Find:

- The frequency, amplitude, wavelength and period of the wave.
- The direction of the wave propagation.
- The velocity of the wave.
- The maximum velocity of the particle.

Answers:

$$\text{a) } f = \frac{\omega}{2\pi} = \frac{2\pi}{2\pi} = 1 \text{ Hz}$$

$$A = 20 \text{ m}$$

$$\lambda = \frac{2\pi}{k} = \frac{2\pi}{0.5\pi} = 4 \text{ m}$$

$$T = \frac{1}{f} = \frac{1}{1} = 1 \text{ s}$$

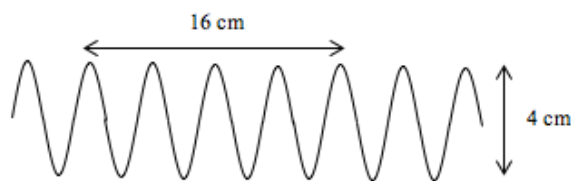
$$\text{b) To the right}$$

$$\text{c) } v = f\lambda = (1)(4) = 4 \text{ m s}^{-1}$$

$$\text{d) } v_{\max} = A\omega = (20)(2\pi) = 125.66 \text{ ms}^{-1}$$

Example 2

Figure below shows a progressive wave travelling at a speed of 24 m s^{-1} to the right.



Determine the wave number

- Calculate the frequency of the wave.
- Write the wave equation.

Answers:

$$\text{a) } \lambda = \frac{16 \text{ cm}}{4} = 4 \text{ cm} = 0.04 \text{ m}$$

$$f = \frac{v}{\lambda} = \frac{24}{0.04} = 600 \text{ Hz}$$

$$\text{b) } A = 2 \text{ cm} = 0.02 \text{ m}$$

$$\omega = 2\pi f = 2\pi(600) = 1200\pi \text{ rad s}^{-1}$$

$$k = \frac{2\pi}{\lambda} = \frac{2\pi}{0.04} = 50\pi \text{ m}^{-1}$$

$$\therefore y = 0.02 \sin(1200\pi t - 50\pi x)$$

where x and y in meter and t is in second.

Example 3

A sinusoidal wave traveling in the $+x$ direction (to the right) has an amplitude of 15.0 cm, a wavelength of 10.0 cm and a frequency of 20.0 Hz.

- Write an expression for the wave function, $y(x,t)$.
- Determine the speed, v_y at $t = 0.5 \text{ s}$ for the particle on the wave located at $x = 5.0 \text{ cm}$.

Answers:

$$\text{a) } y(x,t) = A \sin(\omega t \pm kx)$$

$$A = 15.0 \text{ cm}$$

$$f = 20.0 \text{ Hz}$$

$$\lambda = 10.0 \text{ cm}$$

$$\omega = 2\pi f = 2\pi(20) = 40\pi \text{ rad s}^{-1}$$

$$k = \frac{2\pi}{\lambda} = \frac{2\pi}{10.0} = 0.2\pi \text{ cm}^{-1}$$

$$y(x,t) = 15 \sin(40\pi t - 0.2\pi x)$$

where x and y in centimeter and t is in second.

$$\text{b) } v_y = \frac{dy}{dt}$$

$$v_y = A\omega \cos(\omega t - kx)$$

$$v_y = (15)(40\pi) \cos(40\pi t - 0.2\pi x)$$

$$v_y = 600\pi \cos(40\pi t - 0.2\pi x)$$

$$\text{At } t = 0.5 \text{ s; } x = 5.0 \text{ cm,}$$

$$v_y = 600\pi \cos[40\pi(0.5) - 0.2\pi(5.0)]$$

$$v_y = -600\pi \text{ cm s}^{-1} @ -6\pi \text{ m s}^{-1}$$

Example 4

A progressive wave is represented by the equation

$$y = 5 \sin \left(2\pi t - \frac{\pi}{2} x \right)$$

where y and x are in cm and t is in seconds.

Determine the amplitude, angular frequency, wavelength, period, frequency and wave speed.

Answers:

$$A = 5 \text{ cm}$$

$$\omega = 2\pi \text{ rad s}^{-1}$$

$$\lambda = \frac{2\pi}{k} = \frac{2\pi}{0.5\pi} = 4 \text{ cm}$$

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{2\pi} = 1 \text{ s}$$

$$f = \frac{\omega}{2\pi} = \frac{2\pi}{2\pi} = 1 \text{ Hz}$$

$$v = f\lambda = (1)(4) = 4 \text{ cm s}^{-1}$$

Displacement Graphs of the Wave**Graph of displacement-time (y vs t)**

The graphs show the displacement of all the particles in the wave at any particular distance, x from the origin.

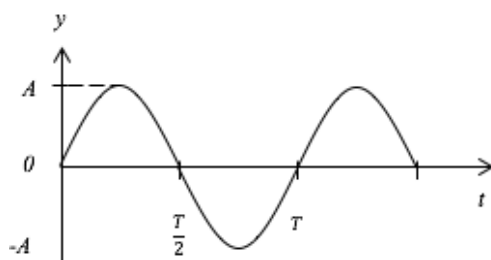
Consider the equation of the wave

$$y = A \sin(\omega t - kx)$$

For the particles at $x = 0$, the equation is given by:

$$y = A \sin[\omega t - k(0)]$$

$$y = A \sin(\omega t)$$



This graph is used to determine:
(Period, Frequency, Angular frequency)

Graph of displacement-distance (y vs x)

The graph shows the displacement of all the particles in the wave at any particular time, t . Consider the equation of the wave

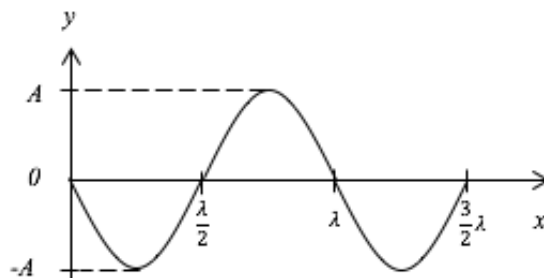
$$y = A \sin(\omega t - kx)$$

For the particle at $t = 0$, the equation of the particle is given by

$$y = A \sin[\omega(0) - kx]$$

$$y = A \sin(-kx)$$

$$y = -A \sin(kx)$$



This graph is used to determine:
(wavelength, wave number)

2.2 Standing Waves

Principle of superposition of waves

Interference is defined as the **interaction (superposition) of two or more wave motions in the same region of space** to produce resultant wave.

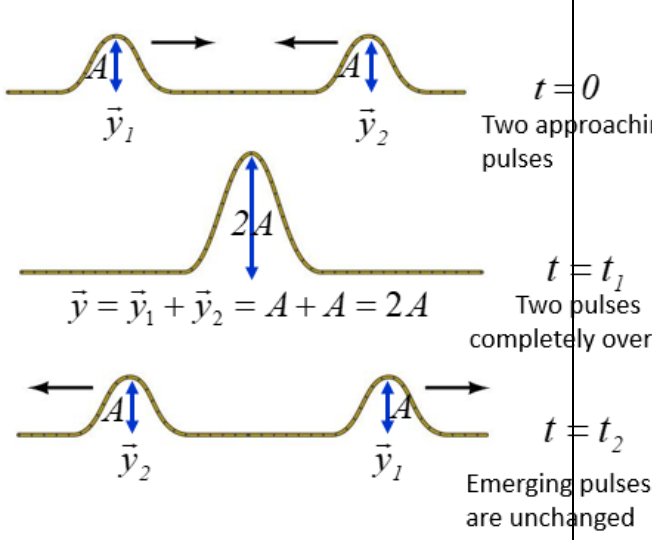
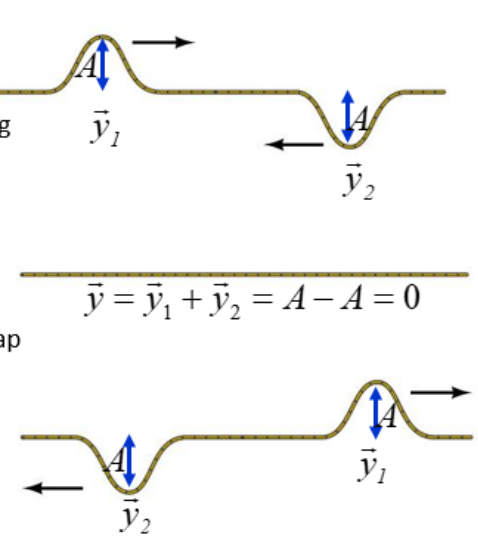
Principle of superposition of waves

When two (or more) waves are moving in the same region, the **resultant displacement at any point is the vector sum of their individual displacement at that point**.

$$\vec{y} = \vec{y}_1 + \vec{y}_2$$

where $\vec{y}_1$ and $\vec{y}_2$ are the displacement of individual pulses at that point.

Constructive interference and destructive interference

Constructive interference	Destructive interference
- occurs when the displacements caused by the two pulses are in the same direction (ie. both are positive/ or negative). - produce a displacement that is larger than the displacement of either of the individual waves.	- occurs when the displacements caused by the two pulses are in opposite directions (ie. one positive, one negative). - produce a displacement that is smaller than the displacements of either of the individual waves.
 $t = 0$ Two approaching pulses $t = t_1$ Two pulses completely overlap $t = t_2$ Emerging pulses are unchanged	 $t = 0$ $t = t_1$ $\vec{y} = \vec{y}_1 + \vec{y}_2 = A - A = 0$

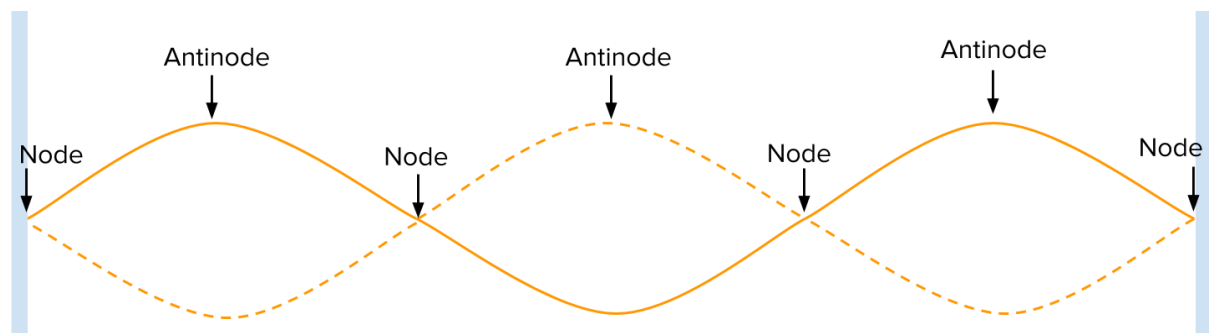
Standing waves (Stationary waves)

Stationary wave is defined as form of wave in which the **profile of the wave does not move** through medium.

It is formed when two progressive waves of same speed, frequency and amplitude which are travelling in opposite directions are superimposed.

In observing the stationary wave, there is no sense of motion in the direction of propagation of either of the original waves (it is called stationary wave because it does not appear to be travelling).

Example of Stationary Wave



Stationary waves in a string

1. When a string is plucked, the progressive wave is produced and travels in both directions along the string.
2. At the end of the string, the waves will be reflected and travel back in the opposite direction. The superposition of incident and reflected waves produce stationary wave with fixed nodes and antinodes as shown in figure above.
3. **Node (N)** is defined as a point at which the **displacement is zero**, where the **destructive interference** occurred.
4. **Antinode (A)** is defined as a point at which the **displacement is maximum** where the **constructive interference** occurred.

Equation of stationary waves

1. Assume two waves with the same amplitude, frequency and wavelength travelling in opposite directions in a medium:

$$y_1 = a \sin(\omega t - kx)$$

and

$$y_2 = a \sin(\omega t + kx)$$

2. Both waves interfere according to the principle of superposition. Therefore,

$$y = y_1 + y_2$$

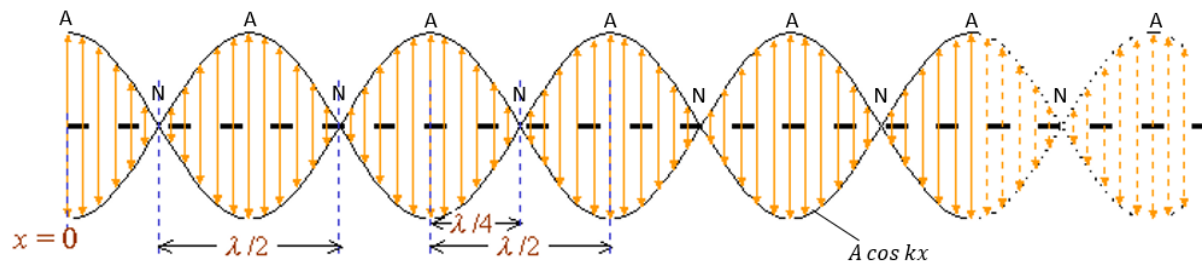
$$y = a \sin(\omega t - kx) + a \sin(\omega t + kx)$$

$$y = 2a \cos kx \sin \omega t$$

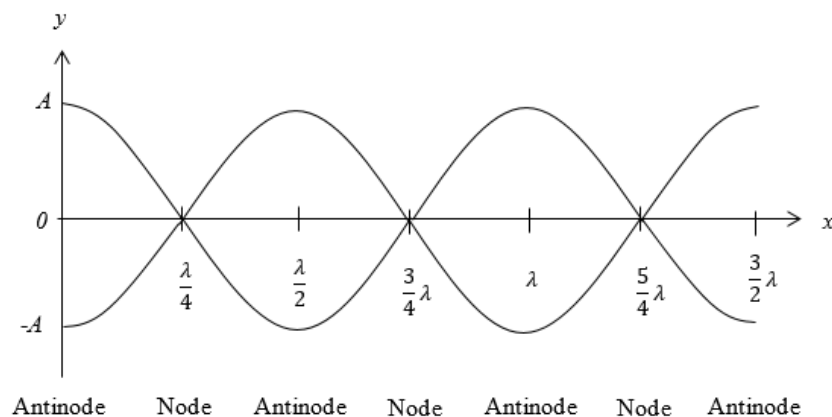
$$y = A \cos kx \sin \omega t$$

There are three types of amplitudes used in describing stationary waves:

- The amplitude of the individual waves, a (**amplitude of progressive wave**)
- The **amplitude** of the simple harmonic motion **of the particles** in the medium.
 - $2a \cos kx = A \cos kx$
 - A given particle in the stationary wave vibrates within the constraints of the envelope function $A \cos kx$
- The **amplitude of stationary wave**, A
 - $A = 2a$



Schematic representation of the standing wave (with an indefinite number of nodes and antinodes) **One starts counting the time ($t = 0$) when the displacement y is maximum at $x = 0$.**



- The distance between adjacent antinodes is $\frac{\lambda}{2}$.
- The distance between adjacent nodes is $\frac{\lambda}{2}$.
- The distance between a node and an adjacent antinode is $\frac{\lambda}{4}$.

Example 5

The expression of a stationary wave is given by

$$y = 0.3 \cos 0.5\pi x \sin 60\pi t$$

Where y and x are in meters and t is in seconds.

- Write the expression for two progressive waves resulting the stationary wave above.
- Determine the wavelength, frequency, amplitude and velocity for both progressive waves.

Answers:

a) $y = A \cos kx \sin \omega t$

By comparison,

$$A = 0.3 \text{ m} \rightarrow a = 0.15 \text{ m}$$

$$k = 0.5\pi \text{ m}^{-1}$$

$$\omega = 60\pi \text{ rad s}^{-1}$$

therefore,

$$y = a \sin(\omega t + kx) \rightarrow y = 0.15 \sin(60\pi t + 0.5\pi x)$$

$$y = a \sin(\omega t - kx) \rightarrow y = 0.15 \sin(60\pi t - 0.5\pi x)$$

where y and x are in meters and t is in seconds.

b) $\lambda = \frac{2\pi}{k} = \frac{2\pi}{0.5\pi} = 4 \text{ m}$

$$f = \frac{\omega}{2\pi} = \frac{60\pi}{2\pi} = 30 \text{ Hz}$$

$$a = \frac{A}{2} = \frac{0.3}{2} = 0.15 \text{ m}$$

$$v = f\lambda = (30)(4)$$

$$v = 120 \text{ m s}^{-1}$$

Example 6

Two harmonic waves are represented by the equations below, where y_1, y_2 and x are in centimetres and t is in seconds.

$$y_1 = 5 \sin(\pi t + 2\pi x)$$

$$y_2 = 5 \sin(\pi t - 2\pi x)$$

- Write an expression for the new wave when both waves are superimposed.
- Determine the wavelength and frequency of stationary waves.

Answers:

a)

$$y_1 = 5 \sin(\pi t + 2\pi x)$$

$$y = a \sin(\omega t + kx)$$

$$y_2 = 5 \sin(\pi t - 2\pi x)$$

$$y = a \sin(\omega t - kx)$$

$$A = 2a = 2(5) = 10$$

$$y = A \cos kx \sin \omega t$$

$$y = 10 \cos 2\pi x \sin \pi t$$

where y and x are in centimeters and t is in seconds.

b) $\lambda = \frac{2\pi}{k} = \frac{2\pi}{2\pi} = 1 \text{ cm}$

$$f = \frac{\omega}{2\pi} = \frac{\pi}{2\pi} = 0.5 \text{ Hz}$$

Comparison between progressive and standing waves

Progressive Wave	Standing Wave
Wave profile move.	Wave profile does not move.
All particles vibrate with the same amplitude.	Particles between two adjacent nodes vibrate with different amplitudes.
Neighbouring particles vibrate with different phases.	Particles between two adjacent nodes vibrate in phase.
All particles vibrate	Particles at nodes do not vibrate at all.
Produced by a disturbance in an elastic medium	Produced by the superposition of two waves moving in opposite direction.
Transmits the energy.	Energy is not transmitted down the string but “stands” in place in the string. (Does not transmit energy)
Simulation:	
	

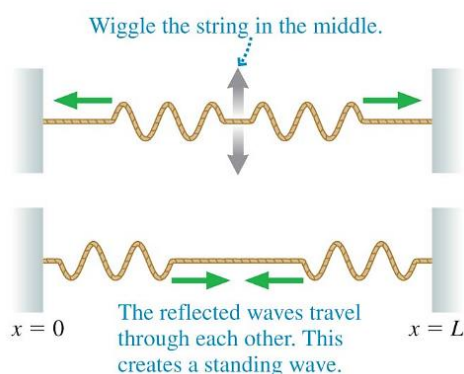
Formation of standing waves in stretched string

When a string of length L stretched between 2 fixed ends is plucked and then released.

Transverse waves produced travel down towards the fixed ends & reflected back.

Wave pulse is being inverted when reflected.

The wave and its reflection wave interfere according to the **superposition** principle and produce standing wave.



Simulation:



Speed of wave on a stretched string is:

$$v = \sqrt{\frac{T}{\mu}}$$

where

v : Speed of wave on a string

T : Tension in the string

μ : Mass per unit length of the string ($\mu = \frac{m}{l}$)

Example 7

A 0.5 kg object is hung at the end of a string of mass 72 g and length 1.2 m . Calculate the speed of the wave propagation in the string when the system is disturbed.

Answers:

$$\mu = \frac{m}{l} = \frac{72 \times 10^{-3}}{1.2} = 0.06 \text{ kg m}^{-1}$$

$$v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{(0.5)(9.81)}{0.06}} = 9.04 \text{ m s}^{-1}$$

Chapter 2 MCQs Practice:**SELECTED FORMULAE**

- | | | |
|--|---------------------------------------|-----------------------------------|
| 1. $x = A \sin \omega t$ | 2. $\omega = 2\pi f = \frac{2\pi}{T}$ | 3. $v = A\omega \cos \omega t$ |
| 4. $a = -A\omega^2 \sin \omega t$ | 5. $k = \frac{2\pi}{\lambda}$ | 6. $v = f\lambda$ |
| 7. $y(x, t) = A \sin(\omega t \pm kx)$ | 8. $v = \frac{dy}{dt}$ | 9. $y = 2A \cos kx \sin \omega t$ |

DIY Quiz

Instruction:

Answer all questions in 10 minutes.

No discussion & do not refer to any notes.

You can only refer to the formulae given below.

- A progressive wave with an amplitude of 2.0 mm , wavelength 25.0 cm and wave speed 5.0 m s^{-1} is traveling in the negative x -axis direction.
 - Determine the wave equation.
 - Calculate the displacement of a particle when $t = 0 \text{ s}$ and $x = 10.0 \text{ cm}$.

Answers:

- $y = 2.0 \times 10^{-3} \sin(40\pi t + 8\pi x)$ where x and y are in meters and t is in seconds.
- $y = 1.18 \times 10^{-3} \text{ m}$

PAST YEAR QUESTIONS

The equation of the progressive wave is given by

$$y(x, t) = 0.250\sin(5\pi t - 6\pi x)$$

Where x is distance in meters, y is displacement in meters and t is time in seconds. Calculate the

- a) Wavelength and frequency.
- b) Speed of wave.
- c) Particle vibrational velocity at $t = 0$ s, $x = 0.100$ m.

[9 marks]